Problem 05: Intelligent Disaster Prediction and Resource Allocation Using Satellite Images, Weather Data, and Geospatial Analytics
Domain: Climate AI | Computer Vision | Time-Series Forecasting | Geospatial ML
Difficulty: Hard

Problem
Floods, wildfires and landslides cause heavy damage when authorities cannot quickly identify vulnerable areas and allocate limited resources. Weather forecasts alone ignore terrain, vegetation, infrastructure, history and population.
Objective
Build a geospatial decision-support platform for ONE disaster type (flood, wildfire or landslide) over a defined region.
Users
Disaster-management authorities, emergency planners.
Core Requirements
- Choose one disaster type and geographic scope.
- Collect historical disaster records and environmental data.
- Process satellite imagery or raster data.
- Extract terrain, vegetation, rainfall features.
- Train a regional risk model.
- Map-based risk visualization.
- Estimate population/infrastructure exposure.
- Rank regions for intervention.
- Explain risk drivers.
- Evaluate with spatial and temporal splits.

Advanced Challenges
- Fusing raster, tabular and time-series data.
- Spatial autocorrelation and geographic leakage.
- Multi-horizon forecasting.
- Rare events.
- Uncertainty in data-sparse areas.
- Resource allocation under a budget (knapsack / LP).

Suggested Datasets
Floods: Sen1Floods11, Global Flood Database, ERA5 rainfall, SRTM/Copernicus DEM, Sentinel-1/2. Wildfire: NASA FIRMS, Next Day Wildfire Spread. Exposure: WorldPop, OpenStreetMap. India options: Bhuvan, IMD, CWC.
Suggested Tech
Python, GeoPandas, Rasterio, XGBoost/LightGBM, PyTorch (U-Net for segmentation), FastAPI, Folium/Plotly, OR-Tools/PuLP.
Reference Architecture
Data ingestion → grid/H3 tiling → feature engineering (terrain, NDVI, rainfall lags) → model (GBM + optional CNN) → uncertainty → exposure join → optimizer (resource allocation) → map dashboard.
Evaluation
PR-AUC, recall on high-risk cells, spatial block cross-validation, temporal hold-out (train past years, test later), allocation effectiveness vs naive rank-by-risk.
Expected Deliverable
Dashboard with risk maps, regional estimates, contributing factors, exposure, priority ranking,

historical validation.
Safety Note
Supports emergency planning; does not guarantee predictions.
Pitfalls
Random split inflates results; adjacent tiles leak; ignoring that “no recorded event” is not “no risk”.
Suggested Timeline (48h)
0-10h data + tiling; 10-24h features + model; 24-34h spatial validation; 34-42h allocation; 42-48h map UI.

Standard Judging Rubric (applies to every problem)

Problem understanding	10%
Technical depth	20%
Data engineering (quality, leakage prevention, reproducibility)	10%
Model performance (metrics, baselines, honest evaluation)	15%
System architecture	15%
Working prototype	10%
Explainability and responsible AI	10%
Innovation and scalability	5%
Presentation and demo	5%

Mandatory Submission Checklist

- □ GitHub repository with setup instructions
- □ Working prototype or recorded demo
- □ Documented system architecture
- □ Dataset sources and preprocessing methodology
- □ Evaluation results with baseline comparisons
- □ Limitations and failure cases
- □ One technically challenging feature demonstrated
- □ Short presentation: problem, approach, results, future work